Practical No. 7

Aim: Write a java program demonstrating abstract class and interface implementation.

Code:

Program 1:

```
// Abstract Parent Class
abstract class Animal {
    private String name;

    public Animal(String name) {
        this.name = name;
    }

    // Abstract method (has no body, MUST be implemented by subclasses)
    public abstract void makeSound();

    // Concrete method (has a body, shared by all subclasses)
    public void sleep() {
        System.out.println(name + " is sleeping... Zzz");
    }

    public String getName() {
        return name;
    }
}

// Concrete Subclass 1
class Dog extends Animal {
    public Dog(String name) {
        super(name);
    }

    // Providing implementation for the abstract method
    @Override
    public void makeSound() {
        System.out.println(getName() + " barks: Woof Woof!");
    }
}

// Concrete Subclass 2
class Cat extends Animal {
    public Cat(String name) {
        super(name);
    }

    // Providing implementation for the abstract method
    @Override
    public void makeSound() {
        System.out.println(getName() + " meows: Meow Meow!");
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        // Animal a = new Animal("Generic"); // ERROR! Cannot instantiate an abstract class.

        // Polymorphism: Using abstract class reference for concrete subclass instances
        Animal myDog = new Dog("Buddy");
        Animal myCat = new Cat("Whiskers");

        myDog.makeSound(); // Outputs: Buddy barks: Woof Woof!
        myDog.sleep();     // Outputs: Buddy is sleeping... Zzz

        System.out.println();

        myCat.makeSound(); // Outputs: Whiskers meows: Meow Meow!
        myCat.sleep();     // Outputs: Whiskers is sleeping... Zzz
    }
}
```

Output:

